

We use the t-test in the Old Faithful example. Here is a brief synopsis of the t-test.

One sample t-test

In the one sample test, given data $x_1, x_2, \dots, x_n$, we test the following hypothesis:

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu \neq \mu_0$$

where μ is the population mean of X and μ_0 is a fixed number. To test the hypothesis, we use the t-statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x} = \sum_i x_i/n$ is the sample mean and $s = \sqrt{\sum_i (x_i - \bar{x})^2/(n-1)}$ is the sample standard deviation that is used as a surrogate for the population standard deviation. We can use a t-distribution with $n-1$ degrees of freedom to estimate the p-value which is the probability of observing a value as extreme as the test result under the null hypothesis. A small p-value provides little support for H_0 and more support for H_1 (we can reject the null hypothesis in this case).

Two sample t-test

In the two sample test, given data $x_1, x_2, \dots, x_{n_1}$ and $y_1, y_2, \dots, y_{n_2}$ we test the following hypothesis:

$$H_0 : \mu_1 - \mu_2 = \mu_0 \text{ versus } H_1 : \mu_1 - \mu_2 \neq \mu_0$$

where μ_1 is the population mean of X , μ_2 is the population mean of Y and μ_0 is a fixed number. To test the hypothesis, we use the t-statistic:

$$t = \frac{\bar{x} - \bar{y} - \mu_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

where $\bar{x} = \sum_i x_i/n_1$ and $\bar{y} = \sum_i y_i/n_2$ are the sample means and $s_1 = \sqrt{\sum_i (x_i - \bar{x})^2/(n_1-1)}$ and $s_2 = \sqrt{\sum_i (y_i - \bar{y})^2/(n_2-1)}$ are the sample standard deviations. The degree of freedom uses a Welch modification which is given by

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$$

We can use a t-distribution with ν degrees of freedom to estimate the p-value which is the probability of observing a value as extreme as the test result under the null hypothesis. The test allows for unequal variances of X and Y and for different sample sizes n_1 and n_2 .